Chapter 7. Principal Component Analysis

Population and sample principal components, SVD, choosing the number of components

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Ch. 1 §6 introduced PCA through sample geometry. Here it is formulated at the population level, with the Lagrange-multiplier derivation, the SVD and the choice of the number of components.

1 Population Principal Components

Let $\mathbf{X} \in \mathbb{R}^p$ with $\mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$, $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$. A linear combination $Z(\mathbf{a}) = \mathbf{a}^\top \mathbf{X}$ has variance $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$. (For $d = 1$: $\mathbf{a}_1 = \arg\max_{\|\mathbf{a}\|=1} \mathrm{Var}(Z(\mathbf{a}))$.)

Definition: principal components

First PC direction vector: $\mathbf{u}_1 = \arg\max_{\|\mathbf{u}\|=1} \mathrm{Var}(Z(\mathbf{u}))$, with $\lambda_1 = \mathrm{Var}(Z_1)$. Second: $\mathbf{u}_2 = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_1 = 0} \mathrm{Var}(\mathbf{u}^\top \mathbf{X})$, with $\lambda_2 = \mathrm{Var}(Z_2)$; and so on:

$$\mathbf{u}_k = \arg \max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_j = 0\ (j<k)} \mathrm{Var}(\mathbf{u}^\top \mathbf{X}), \qquad Z_k = \mathbf{u}_k^\top \mathbf{X}.$$

Then $\lambda_1 \geq \lambda_2 \geq \cdots$ and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Theorem: solution

$\mathbf{u}_k$ is the kth eigenvector of $\boldsymbol{\Sigma}$, $\mathrm{Var}(Z_k) = \lambda_k$, and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Lagrange multipliers (handwritten notes). **1.** $\Phi(\mathbf{u}, \lambda) = \mathrm{Var}(\mathbf{u}^\top \mathbf{X}) - \lambda(\mathbf{u}^\top \mathbf{u} - 1) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1)$. $\frac{1}{2} \frac{\partial}{\partial \mathbf{u}} \Phi(\mathbf{u}, \lambda) = \boldsymbol{\Sigma}\mathbf{u} - \lambda \mathbf{u} = \mathbf{0}$, so $\boldsymbol{\Sigma}\mathbf{u} = \lambda \mathbf{u}$ and $\lambda = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u}$: the objective equals the eigenvalue, so take the largest, λ_1.

2. With $\boldsymbol{\gamma}^k = (\gamma_1, \dots, \gamma_{k-1})$,

$$\Phi(\mathbf{u}, \lambda, \boldsymbol{\gamma}^k) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1) - \sum_{j=1}^{k-1} 2\gamma_j \mathbf{u}_j^\top \mathbf{u}, \qquad \frac{1}{2} \frac{\partial \Phi}{\partial \mathbf{u}} = \boldsymbol{\Sigma}\mathbf{u} - \lambda \mathbf{u} - \sum_{j=1}^{k-1} \gamma_j \mathbf{u}_j = \mathbf{0}, \quad \frac{\partial \Phi}{\partial \gamma_j} = \mathbf{u}_j^\top \mathbf{u} = 0.$$

Premultiplying by $\mathbf{u}_i^\top$ $(i < k)$: $\mathbf{u}_i^\top \boldsymbol{\Sigma} \mathbf{u} = \gamma_i$, and $\mathbf{u}_i^\top \boldsymbol{\Sigma} \mathbf{u} = \lambda_i \mathbf{u}_i^\top \mathbf{u} = 0$, so $\gamma_i = 0$. Hence $\boldsymbol{\Sigma}\mathbf{u} = \lambda \mathbf{u}$ again, and among eigenvectors orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$ the largest eigenvalue λ_k is chosen: $\mathbf{u}_k, \lambda_k$. Finally $\mathrm{Cov}(Z_j, Z_k) = \mathbf{u}_j^\top \boldsymbol{\Sigma} \mathbf{u}_k = \lambda_k \mathbf{u}_j^\top \mathbf{u}_k = 0$. ■

Theorem: total variance and correlations

1. $\sum_{j=1}^p \mathrm{Var}(X_j) = \mathrm{tr}\boldsymbol{\Sigma} = \sum_{k=1}^p \lambda_k = \sum_k \mathrm{Var}(Z_k)$; the proportion of variance explained by PC k is $\lambda_k / \sum_j \lambda_j$.
2. $\mathrm{Corr}(Z_k, X_j) = \dfrac{u_{jk}\sqrt{\lambda_k}}{\sqrt{\sigma_{jj}}}$, where u_{jk} is the jth entry of $\mathbf{u}_k$.

Proof. (1) $\mathrm{tr}(\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top) = \mathrm{tr}\boldsymbol{\Lambda}$. (2) $\mathrm{Cov}(Z_k, X_j) = \mathbf{u}_k^\top \boldsymbol{\Sigma} \mathbf{e}_j = \lambda_k \mathbf{u}_k^\top \mathbf{e}_j = \lambda_k u_{jk}$, so $\mathrm{Corr} = \lambda_k u_{jk} / (\sqrt{\lambda_k}\sqrt{\sigma_{jj}})$. ■

Theorem: Ky Fan maximum principle — q-dimensional subspaces

$$\max_{\mathbf{V} \in \mathbb{R}^{p \times q},\ \mathbf{V}^\top \mathbf{V} = \mathbf{I}_q} \mathrm{tr}(\mathbf{V}^\top \boldsymbol{\Sigma} \mathbf{V}) = \lambda_1 + \cdots + \lambda_q,$$

attained at $\mathbf{V} = [\mathbf{u}_1, \dots, \mathbf{u}_q]$ (and its rotations).

Proof. $\mathbf{P} = \mathbf{V}\mathbf{V}^\top$ is a rank-q projection with eigenvalues 1 (q times) and 0. By von Neumann's upper bound, $\operatorname{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V}) = \operatorname{tr}(\boldsymbol{\Sigma}\mathbf{P}) \le \sum_i \lambda_i(\boldsymbol{\Sigma})\lambda_i(\mathbf{P}) = \lambda_1 + \cdots + \lambda_q$. ■

Remark: PCA as optimal approximation

$\mathrm{E}\|(\mathbf{X}-\boldsymbol{\mu}) - \mathbf{V}\mathbf{V}^\top(\mathbf{X}-\boldsymbol{\mu})\|^2 = \operatorname{tr}\boldsymbol{\Sigma} - \operatorname{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V})$, so among q-dimensional subspaces through the mean, the first q PCs minimise the expected squared reconstruction error, the minimum being $\lambda_{q+1} + \cdots + \lambda_p$.

2 Sample Principal Components and the SVD

2.1 Sample PCs

Let $\tilde{\mathbf{X}} = \mathbf{H}_n\mathbf{X} = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top$ be the centred data matrix, $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\boldsymbol{\Lambda}}\hat{\mathbf{U}}^\top$, and

$$\hat{\mathbf{u}}_k = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top\hat{\mathbf{u}}_j=0,\ j=1,\dots,k-1} \mathbf{u}^\top\mathbf{S}\mathbf{u}, \qquad \text{scores } t_{ik} = \hat{\mathbf{u}}_k^\top(\mathbf{x}_i - \bar{\mathbf{x}}).$$

The sample PC scores have sample variance $\hat{\lambda}_k$ and are mutually uncorrelated.

2.2 Relation to the singular value decomposition

Write the SVD of the $p \times n$ matrix $\tilde{\mathbf{X}}^\top$ ($n > p$) as

$$\tilde{\mathbf{X}}^\top = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \underbrace{[\mathbf{u}_1 \cdots \mathbf{u}_p]}_{p\times p} \underbrace{\begin{pmatrix} d_1 & & \\ & \ddots & \\ & & d_p \end{pmatrix}}_{p\times p} \underbrace{\begin{pmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_p^\top \end{pmatrix}}_{p\times n}, \qquad \mathbf{U}^\top\mathbf{U} = \mathbf{I}_p,\ \mathbf{V}^\top\mathbf{V} = \mathbf{I}_p,\ d_1 \ge \cdots \ge d_p \ge 0.$$

Then

$$\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \mathbf{U}\operatorname{diag}\Big(\frac{d_1^2}{n-1}, \dots, \frac{d_p^2}{n-1}\Big)\mathbf{U}^\top, \qquad \begin{pmatrix} \mathbf{z}_{(1)}^\top \\ \vdots \\ \mathbf{z}_{(p)}^\top \end{pmatrix} = \mathbf{Z} = \mathbf{U}^\top\tilde{\mathbf{X}}^\top = \mathbf{D}\mathbf{V}^\top = \begin{pmatrix} d_1\mathbf{v}_1^\top \\ \vdots \\ d_p\mathbf{v}_p^\top \end{pmatrix}.$$

The PC directions are the left singular vectors, the eigenvalues are $d_k^2/(n-1)$, and the kth PC score vector is $\mathbf{z}_{(k)} = \hat{\mathbf{u}}_k^\top\tilde{\mathbf{X}}^\top = d_k\mathbf{v}_k^\top$. Numerically the SVD is preferred to forming $\mathbf{S}$ (computation of PCA).

3 PCA and the Best Approximating Affine Subspace

Theorem: claim

PCA with r components is equivalent to finding the best r-dimensional affine subspace (hyperplane) approximating the data.

3.1 The problem

Let the data be $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ and $r \in \{1, \dots, p\}$. We wish to approximate $\mathbf{x}_1, \dots, \mathbf{x}_n$ by $\bar{\mathbf{x}} + \mathbf{a}_1, \dots, \bar{\mathbf{x}} + \mathbf{a}_n$ where $\operatorname{rank}([\mathbf{a}_1, \dots, \mathbf{a}_n]) = r$. The approximation of $\mathbf{x}_i$ by $\bar{\mathbf{x}} + \mathbf{a}_i$ is better when the

error $\|\mathbf{x}_i - (\bar{\mathbf{x}} + \mathbf{a}_i)\|_2^2$ is smaller, and the approximation of $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ by $\{\bar{\mathbf{x}} + \mathbf{a}_i\}$ is better when $\mathrm{SSE} = \sum_{i=1}^n \|\mathbf{x}_i - (\bar{\mathbf{x}} + \mathbf{a}_i)\|_2^2$ is smaller. That is, we wish to find $\mathbf{A}_r = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ $(p \times n)$ solving

$$(\star) \qquad \min_{\mathbf{A}_r} \sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{a}_i\|_2^2 \quad \text{subject to} \quad \operatorname{rank}(\mathbf{A}_r) = r.$$

Since every column of a rank-r matrix lies in an r-dimensional subspace, the points $\bar{\mathbf{x}} + \mathbf{a}_i$ lie on an r-dimensional affine subspace through $\bar{\mathbf{x}}$; conversely any such subspace gives feasible $\mathbf{A}_r$.

3.2 Observations

1. With $\tilde{\mathbf{X}}^\top = [\mathbf{x}_1 - \bar{\mathbf{x}}, \dots, \mathbf{x}_n - \bar{\mathbf{x}}]$ $(p \times n)$,

$$\sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{a}_i\|_2^2 = \|\tilde{\mathbf{X}}^\top - \mathbf{A}_r\|_F^2,$$

because for a $p \times n$ matrix $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_n]$ the Frobenius norm is $\|\mathbf{B}\|_F^2 = \sum_{i=1}^p \sum_{j=1}^n b_{ij}^2 = \sum_{j=1}^n \|\mathbf{b}_j\|_2^2 = \operatorname{tr}(\mathbf{B}\mathbf{B}^\top) = \operatorname{tr}(\mathbf{B}^\top\mathbf{B})$.

2. The sample PCA of $(\mathbf{x}_1, \dots, \mathbf{x}_n)$ is given by

   (1) the eigendecomposition $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top = \sum_{i=1}^p \hat{\lambda}_i \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{\lambda}_i$ the ith PC variance, $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_r > \hat{\lambda}_{r+1} \geq \cdots \geq \hat{\lambda}_p$;

   (2) the SVD $\tilde{\mathbf{X}}_{(p\times n)}^\top = \hat{\mathbf{U}}\hat{\mathbf{D}}\hat{\mathbf{V}}^\top = \sum_{i=1}^p \hat{d}_i \hat{\mathbf{u}}_i \hat{\mathbf{v}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{d}_i\hat{\mathbf{v}}_i$ the ith PC score vector;

   (3) $\hat{\mathbf{u}}_i^\top \tilde{\mathbf{X}}^\top = \hat{d}_i \hat{\mathbf{v}}_i^\top$, i.e.

$$\begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix}.$$

Theorem: solution (Eckart–Young–Mirsky)

The $\mathbf{A}_r$ that minimises $(\star)$ is

$$\mathbf{A}_r = \sum_{i=1}^r \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r][\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r]^\top \tilde{\mathbf{X}}^\top,$$

the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace, and the minimum is $\sum_{i>r} \hat{d}_i^2 = (n-1)\sum_{i=r+1}^p \hat{\lambda}_i$.

Proof. Step 1 (parametrise rank-r matrices). Since $\operatorname{rank}(\mathbf{A}_r) = r$, we can write $\mathbf{A}_r = \mathbf{E}\mathbf{C}^\top$ with $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_r]$ $(p \times r)$ satisfying $\mathbf{E}^\top\mathbf{E} = \mathbf{I}_r$ (an orthonormal basis of the column space of $\mathbf{A}_r$) and $\mathbf{C} = \begin{pmatrix} \mathbf{c}_1^\top \\ \vdots \\ \mathbf{c}_n^\top \end{pmatrix}$ an $n \times r$ matrix. Writing $\tilde{\mathbf{x}}_i = \mathbf{x}_i - \bar{\mathbf{x}}$, $(\star)$ is equivalent to $\min_{\mathbf{E},\mathbf{C}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2$, where

$$(\star\star) \qquad \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2 = \big\|(\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_n) - \mathbf{E}[\mathbf{c}_1, \dots, \mathbf{c}_n]\big\|_F^2 = \sum_{i=1}^n \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2.$$

Step 2 (optimal $\mathbf{C}$ for fixed $\mathbf{E}$). For any given $\mathbf{E}$, $(\star\star)$ separates over i and its minimiser is

$$\hat{\mathbf{c}}_i = \arg\min_{\mathbf{c}_i} \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2 = (\mathbf{E}^\top\mathbf{E})^{-1}\mathbf{E}^\top\tilde{\mathbf{x}}_i = \mathbf{E}^\top\tilde{\mathbf{x}}_i,$$

the projection scores of $\tilde{\mathbf{x}}_i$ onto span($\mathbf{E}$) (least squares). Stacking, $\hat{\mathbf{C}} = \begin{pmatrix} \tilde{\mathbf{x}}_1^\top \mathbf{E} \\ \vdots \\ \tilde{\mathbf{x}}_n^\top \mathbf{E} \end{pmatrix} = \tilde{\mathbf{X}}\mathbf{E}$, i.e. $\hat{\mathbf{C}}^\top = \mathbf{E}^\top \tilde{\mathbf{X}}^\top$.

Step 3 (profile out C). Plugging $\hat{\mathbf{C}}$ into ($\star\star$) (profiling, Prop. 6.3), and using that $\mathbf{I} - \mathbf{E}\mathbf{E}^\top$ is symmetric idempotent, $(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)^2 = \mathbf{I} - \mathbf{E}\mathbf{E}^\top$,

$$\begin{aligned} \min_{\mathbf{E}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{E}^\top \tilde{\mathbf{X}}^\top\|_F^2 &= \min_{\mathbf{E}} \|(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top\|_F^2 = \min_{\mathbf{E}} \operatorname{tr}(\tilde{\mathbf{X}}(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top) \\ &= \min_{\mathbf{E}} \left\{ \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1)\operatorname{tr}\Big(\mathbf{E}^\top \frac{\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}}{n-1} \mathbf{E}\Big) \right\} \\ &= \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1) \max_{\mathbf{E}} \operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top). \end{aligned}$$

Step 4 (von Neumann). Recall $\mathbf{S} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top$. $\mathbf{E}\mathbf{E}^\top$ is symmetric with eigenvalues 1 (r times) and 0 ($p - r$ times), so von Neumann's trace inequality (Lemma 3 of the lecture: $|\operatorname{tr}(\mathbf{A}\mathbf{B}^\top)| \le \boldsymbol{\sigma}_A^\top \boldsymbol{\sigma}_B$, equality when $\mathbf{A}$ and $\mathbf{B}$ have the same eigenvectors) gives

$$\operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top) \le \hat{\lambda}_1 + \cdots + \hat{\lambda}_r,$$

with equality when $\mathbf{E} = [\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_r]$ (or any $\mathbf{E}$ spanning the same space).

Step 5 (the minimiser). Therefore

$$\mathbf{A}_r = \mathbf{E}\hat{\mathbf{C}}^\top = [\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = [\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix} = \sum_{i=1}^{r} \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top$$

by Observation 2(3): the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace. Since $\operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) = (n-1)\operatorname{tr}\mathbf{S} = (n-1)\sum_i \hat{\lambda}_i$, the minimum is $(n-1)\sum_{i>r} \hat{\lambda}_i = \sum_{i>r} \hat{d}_i^2$. ■

Remark: interpretation

- The best approximating hyperplane is $\bar{\mathbf{x}} + \text{span}\{\hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_r\}$ and each $\mathbf{x}_i$ is approximated by its orthogonal projection $\bar{\mathbf{x}} + \sum_{k \le r} (\hat{\mathbf{u}}_k^\top \tilde{\mathbf{x}}_i)\hat{\mathbf{u}}_k$: the PC scores are the coordinates on the hyperplane.
- The fitted values satisfy $\hat{\mathbf{c}}_i = \mathbf{E}^\top \tilde{\mathbf{x}}_i$, i.e. reconstruction uses projection scores, exactly the decomposition $\mathbf{x} = \bar{\mathbf{x}} + \sum s_{\mathbf{v}_j}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_j + r_{\mathcal{L}_q}(\mathbf{x})$ of Ch. 1.
- Minimising squared error ("PCA and approximation plane") and maximising retained variance (the $\arg\max \mathbf{u}^\top \mathbf{S}\mathbf{u}$ definition) are the same problem, since SSE $= (n-1)\{\operatorname{tr}\mathbf{S} - \operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top)\}$.
- If $\hat{\lambda}_r = \hat{\lambda}_{r+1}$ the optimal subspace is not unique; the strict gap $\hat{\lambda}_r > \hat{\lambda}_{r+1}$ makes it unique.

3.3 Correlation PCA

If the units differ, use PCA of standardised data: $\mathbf{R} = \mathbf{D}_{\mathbf{\Sigma}}^{-1/2} \mathbf{\Sigma} \mathbf{D}_{\mathbf{\Sigma}}^{-1/2} = \mathbf{U}_R \mathbf{\Lambda}_R \mathbf{U}_R^\top$ (sample: $\mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2}$). Since $\operatorname{tr}\mathbf{R} = p$, the proportion explained is λ_k^R / p. Covariance and correlation PCs are in general not simple transformations of each other (scale dependence).

4 How Many Components to Keep?

- **Scree plot / cumulative scree plot**: plot $\hat{\lambda}_k$ against k and look for the elbow, or take the q at which the cumulative proportion reaches a target (e.g. 80–90%).
- **Kaiser's rule**: keep components with $\hat{\lambda}_k > \bar{\lambda} = \frac{1}{p}\sum_j \hat{\lambda}_j$; for correlation PCA, $\hat{\lambda}_k > 1$.
- **Likelihood ratio testing**: under normality test $H_0(k) : \lambda_{k+1} = \cdots = \lambda_p$ (the remaining space is spherical). With arithmetic mean $\bar{a}$ and geometric mean $\bar{g}$ of the last $p-k$ sample eigenvalues,

$$-2\log\Lambda = n(p-k)\log\frac{\bar{a}}{\bar{g}} \;\dot{\sim}\; \chi^2_{(p-k+2)(p-k-1)/2};$$

since $\bar{a} \geq \bar{g}$ (AM–GM), the statistic is small when the eigenvalues are nearly equal.

Remark: cautions in interpretation

- The sign of an eigenvector is arbitrary; what matters is the sign pattern (same signs: "size"; mixed signs: "contrast"). In the scores example $\mathbf{u}_1 \propto (1,1)$ was overall level and $\mathbf{u}_2 \propto (1,-1)$ the contrast between subjects.
- With tied eigenvalues the directions within the eigenspace are not unique.
- Sample eigenvalues are biased: large ones are overestimated and small ones underestimated. Under normality, for distinct λ_k, $\sqrt{n}(\hat{\lambda}_k - \lambda_k) \Rightarrow N(0, 2\lambda_k^2)$.
- High explained variance does not mean usefulness for classification or prediction (the $X_1 = 10Z$, $X_2 = Y$ example of Ch. 1).

5 Example: Scores in Two Subjects Revisited

For $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$ the characteristic equation $(125-\lambda)^2 - 75^2 = 0$ gives $\lambda = 200, 50$; the null space of $\mathbf{S} - 200\mathbf{I} = \begin{pmatrix} -75 & 75 \\ 75 & -75 \end{pmatrix}$ is spanned by $(1,1)^\top$. $\mathrm{Corr}(Z_1, X_1) = u_{11}\sqrt{\lambda_1}/\sqrt{s_{11}} = \frac{1}{\sqrt{2}}\sqrt{200}/\sqrt{125} = 0.894$, $\mathrm{Corr}(Z_2, X_1) = \frac{1}{\sqrt{2}}\sqrt{50}/\sqrt{125} = 0.447$, and $0.894^2 + 0.447^2 = 1$: each variable's variance is fully decomposed over the PCs.

Chapter 8. Canonical Correlation Analysis

Maximal correlation between two sets of variables

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1 Setting

We want to summarise the association between two sets of variables $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$ with $\operatorname{Cov}\begin{pmatrix}\mathbf{X}\\ \mathbf{Y}\end{pmatrix} = \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix}$, $\boldsymbol{\Sigma}_{11}, \boldsymbol{\Sigma}_{22} \succ 0$. Instead of inspecting pq correlations, maximise the correlation of linear combinations

$$\xi = \mathbf{g}^\top \mathbf{X}, \qquad \omega = \mathbf{h}^\top \mathbf{Y}.$$

For $q = 1$ this is the multiple correlation (Ch. 3). PCA maximises variance *within* one set; CCA maximises correlation *between* two sets.

Definition: canonical variates and canonical correlations

$$(\mathbf{g}_1, \mathbf{h}_1) = \arg \max_{\mathbf{g}\in\mathbb{R}^p,\, \mathbf{h}\in\mathbb{R}^q} \operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}),$$

$$(\mathbf{g}_k, \mathbf{h}_k) = \arg \max_{\substack{\mathbf{g}\in\mathbb{R}^p,\, \mathbf{h}\in\mathbb{R}^q \\ \mathbf{g}^\top \boldsymbol{\Sigma}_{11} \mathbf{g}_j = 0,\ \mathbf{h}^\top \boldsymbol{\Sigma}_{22} \mathbf{h}_j = 0,\ j = 1,\dots,k-1}} \operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}).$$

$\xi_k = \mathbf{g}_k^\top \mathbf{X}$, $\omega_k = \mathbf{h}_k^\top \mathbf{Y}$ form the kth pair of canonical variates and $\rho_k = \operatorname{Corr}(\xi_k, \omega_k)$ is the kth canonical correlation. Correlation is scale-free, so normalise $\operatorname{Var}(\xi_k) = \operatorname{Var}(\omega_k) = 1$. The constraints mean $\operatorname{Cov}(\xi_k, \xi_j) = \operatorname{Cov}(\omega_k, \omega_j) = 0$, and $\rho_k \le \rho_j$ for $j < k$.

$\rho(\mathbf{X}, \mathbf{Y}) := \max \operatorname{Corr}(\mathbf{a}^\top \mathbf{X}, \mathbf{b}^\top \mathbf{Y}) = \rho_1$ is the **largest canonical coefficient**.

2 Solution via the Singular Value Decomposition

Theorem: solution of CCA

Let $\mathbf{M} = \boldsymbol{\Sigma}_{11}^{-1/2} \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1/2}$ $(p \times q)$ have SVD $\mathbf{M} = \sum_{k=1}^{r} d_k \mathbf{a}_k \mathbf{b}_k^\top$, $d_1 \ge \cdots \ge d_r > 0$, $r = \operatorname{rank} \boldsymbol{\Sigma}_{12}$. Then

$$\rho_k = d_k, \qquad \mathbf{g}_k = \boldsymbol{\Sigma}_{11}^{-1/2} \mathbf{a}_k, \qquad \mathbf{h}_k = \boldsymbol{\Sigma}_{22}^{-1/2} \mathbf{b}_k.$$

Equivalently $\mathbf{a}_k$ is the kth eigenvector of $\mathbf{M}\mathbf{M}^\top = \boldsymbol{\Sigma}_{11}^{-1/2} \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1/2}$, $\mathbf{b}_k$ that of $\mathbf{M}^\top \mathbf{M} = \boldsymbol{\Sigma}_{22}^{-1/2} \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1/2}$, and ρ_k^2 is the eigenvalue.

Proof. Substituting $\mathbf{a} = \boldsymbol{\Sigma}_{11}^{1/2} \mathbf{g}$, $\mathbf{b} = \boldsymbol{\Sigma}_{22}^{1/2} \mathbf{h}$,

$$\operatorname{Corr}(\mathbf{g}^\top \mathbf{X}, \mathbf{h}^\top \mathbf{Y}) = \frac{\mathbf{g}^\top \boldsymbol{\Sigma}_{12} \mathbf{h}}{\sqrt{\mathbf{g}^\top \boldsymbol{\Sigma}_{11} \mathbf{g}} \sqrt{\mathbf{h}^\top \boldsymbol{\Sigma}_{22} \mathbf{h}}} = \frac{\mathbf{a}^\top \mathbf{M} \mathbf{b}}{\sqrt{\mathbf{a}^\top \mathbf{a}} \sqrt{\mathbf{b}^\top \mathbf{b}}},$$

so we maximise $\mathbf{a}^\top \mathbf{M} \mathbf{b}$ over $\|\mathbf{a}\| = \|\mathbf{b}\| = 1$. With $\mathbf{M} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$, $\boldsymbol{\alpha} = \mathbf{U}^\top \mathbf{a}$, $\boldsymbol{\beta} = \mathbf{V}^\top \mathbf{b}$, Cauchy–Schwarz gives

$$\mathbf{a}^\top \mathbf{M} \mathbf{b} = \mathbf{a}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top \mathbf{b} = \sum_k d_k \alpha_k \beta_k \le d_1 \sum_k |\alpha_k \beta_k| \le d_1 \|\boldsymbol{\alpha}\| \|\boldsymbol{\beta}\| \le d_1,$$

with equality at $\mathbf{a} = \mathbf{a}_1$, $\mathbf{b} = \mathbf{b}_1$. Under $\mathbf{g}^\top \boldsymbol{\Sigma}_{11} \mathbf{g}_j = \mathbf{a}^\top \mathbf{a}_j = 0$ and $\mathbf{b}^\top \mathbf{b}_j = 0$ $(j < k)$ we have $\alpha_j = \beta_j = 0$ for $j < k$, and the same argument gives maximum d_k at $(\mathbf{a}_k, \mathbf{b}_k)$. ∎

Theorem: properties of canonical variates

1. $\mathrm{Var}(\xi_k) = \mathrm{Var}(\omega_k) = 1$ and $\mathrm{Corr}(\xi_k, \xi_l) = \mathrm{Corr}(\omega_k, \omega_l) = 0$ for $k \neq l$.
2. $\mathrm{Corr}(\xi_k, \omega_l) = 0$ for $k \neq l$, $\mathrm{Corr}(\xi_k, \omega_k) = \rho_k$.
3. Invariance: canonical correlations are unchanged by $\mathbf{X} \mapsto \mathbf{D}_1\mathbf{X} + \mathbf{c}_1$, $\mathbf{Y} \mapsto \mathbf{D}_2\mathbf{Y} + \mathbf{c}_2$ with $\mathbf{D}_i$ non-singular; covariance and correlation matrices give the same ρ_k.
4. For $q = 1$, $\rho_1^2 = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}/\sigma_{22}$ is the squared multiple correlation.

Proof. (1) $\mathrm{Cov}(\xi_k, \xi_l) = \mathbf{g}_k^\top \boldsymbol{\Sigma}_{11}\mathbf{g}_l = \mathbf{a}_k^\top \mathbf{a}_l = \delta_{kl}$. (2) $\mathrm{Cov}(\xi_k, \omega_l) = \mathbf{g}_k^\top \boldsymbol{\Sigma}_{12}\mathbf{h}_l = \mathbf{a}_k^\top \mathbf{M}\mathbf{b}_l = d_l \mathbf{a}_k^\top \mathbf{a}_l = d_l\delta_{kl}$. (3) $\mathrm{Corr}(\mathbf{g}^\top\mathbf{D}_1\mathbf{X}, \mathbf{h}^\top\mathbf{D}_2\mathbf{Y}) = \mathrm{Corr}((\mathbf{D}_1^\top\mathbf{g})^\top\mathbf{X}, (\mathbf{D}_2^\top\mathbf{h})^\top\mathbf{Y})$ and $\mathbf{g} \mapsto \mathbf{D}_1^\top\mathbf{g}$ is a bijection, so the maxima agree. (4) For $q = 1$ the only nonzero eigenvalue of $\mathbf{M}\mathbf{M}^\top$ is $\mathbf{M}^\top\mathbf{M} = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}/\sigma_{22}$. ■

3 Sample Canonical Correlations

Replace the covariances by $\mathbf{S}_{11} = \widehat{\mathrm{Cov}}(\mathbf{X})$, $\mathbf{S}_{22} = \widehat{\mathrm{Cov}}(\mathbf{Y})$, $\mathbf{S}_{12} = \widehat{\mathrm{Cov}}(\mathbf{X}, \mathbf{Y})$:

$$(\hat{\mathbf{g}}_k, \hat{\mathbf{h}}_k) = \arg \max_{\substack{\mathbf{g}\in\mathbb{R}^p,\, \mathbf{h}\in\mathbb{R}^q \\ \mathbf{g}^\top\mathbf{S}_{11}\hat{\mathbf{g}}_j = 0,\ \mathbf{h}^\top\mathbf{S}_{22}\hat{\mathbf{h}}_j = 0,\ j=1,\dots,k-1}} \frac{\mathbf{g}^\top\mathbf{S}_{12}\mathbf{h}}{\sqrt{\mathbf{g}^\top\mathbf{S}_{11}\mathbf{g}\,\mathbf{h}^\top\mathbf{S}_{22}\mathbf{h}}}.$$

With $\mathbf{a} = \mathbf{S}_{11}^{1/2}\mathbf{g}$, $\mathbf{b} = \mathbf{S}_{22}^{1/2}\mathbf{h}$ this is maximising $\dfrac{\mathbf{a}^\top\mathbf{S}_{11}^{-1/2}\mathbf{S}_{12}\mathbf{S}_{22}^{-1/2}\mathbf{b}}{\sqrt{\mathbf{a}^\top\mathbf{a}\,\mathbf{b}^\top\mathbf{b}}} = \max_{\|\mathbf{a}\|=\|\mathbf{b}\|=1} \mathbf{a}^\top\mathbf{M}\mathbf{b} = \mathbf{a}^\top\mathbf{U}\mathbf{D}\mathbf{V}^\top\mathbf{b}$, so

$$\mathbf{a}_i = \mathbf{v}_i(\mathbf{S}_{11}^{-1/2}\mathbf{S}_{12}\mathbf{S}_{22}^{-1}\mathbf{S}_{21}\mathbf{S}_{11}^{-1/2}), \qquad \mathbf{b}_i = \mathbf{v}_i(\mathbf{S}_{22}^{-1/2}\mathbf{S}_{21}\mathbf{S}_{11}^{-1}\mathbf{S}_{12}\mathbf{S}_{22}^{-1/2}),$$

where $\mathbf{v}_i(\mathbf{M})$ is the ith eigenvector of $\mathbf{M}$.

3.1 Regularised CCA

If p or q is large relative to n, $\mathbf{S}_{11}, \mathbf{S}_{22}$ are singular or unstable and sample canonical correlations are biased towards 1. **Regularised CCA** adds a ridge term,

$$\mathbf{S}_{11} \mapsto \mathbf{S}_{11} + \lambda_1\mathbf{I}, \qquad \mathbf{S}_{22} \mapsto \mathbf{S}_{22} + \lambda_2\mathbf{I},$$

with λ_1, λ_2 chosen by cross-validation.

4 Testing Canonical Correlations

Under multivariate normality the likelihood ratio statistic for $H_0 : \boldsymbol{\Sigma}_{12} = \mathbf{0}$ (all $\rho_k = 0$) is

$$\Lambda^{2/n} = \frac{|\mathbf{S}|}{|\mathbf{S}_{11}||\mathbf{S}_{22}|} = \prod_{k=1}^{\min(p,q)} (1 - \hat{\rho}_k^2),$$

and Bartlett's approximation gives $-\big(n - 1 - \tfrac{p+q+1}{2}\big)\sum_k \ln(1-\hat{\rho}_k^2) \,\dot\sim\, \chi^2_{pq}$. The hypothesis that only the first k are nonzero, $H_0^{(k)} : \rho_{k+1} = \cdots = 0$, is tested sequentially with

$$-\Big(n - 1 - \frac{p+q+1}{2}\Big) \sum_{i=k+1}^{\min(p,q)} \ln(1-\hat{\rho}_i^2) \;\dot\sim\; \chi^2_{(p-k)(q-k)}.$$

Derivation: the determinant identity

By the block determinant formula, $|\mathbf{S}| = |\mathbf{S}_{22}||\mathbf{S}_{11} - \mathbf{S}_{12}\mathbf{S}_{22}^{-1}\mathbf{S}_{21}| = |\mathbf{S}_{22}||\mathbf{S}_{11}|\,|\mathbf{I} - \hat{\mathbf{M}}\hat{\mathbf{M}}^\top|$, and the eigenvalues of $\hat{\mathbf{M}}\hat{\mathbf{M}}^\top$ are $\hat{\rho}_k^2$, so $|\mathbf{S}|/(|\mathbf{S}_{11}||\mathbf{S}_{22}|) = \prod(1 - \hat{\rho}_k^2)$.

5 Relations to PCA, Multiple Correlation and Discriminant Analysis

- **Multiple correlation**: CCA with $q = 1$.
- **Discriminant analysis**: with $\mathbf{Y}$ a set of group indicator (dummy) variables, the canonical variates of $\mathbf{X}$ give the same directions as Fisher's discriminants (Ch. 10).
- **PCA**: variance maximisation within one set (eigendecomposition of $\boldsymbol{\Sigma}$) versus correlation maximisation between two sets (SVD of $\boldsymbol{\Sigma}_{11}^{-1/2}\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1/2}$).

Chapter 9. Factor Analysis

Model, estimation (principal components, principal factors, MLE), rotation, factor scores

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1 The Orthogonal Factor Model

The covariance of p observed variables is explained by a few, k, unobserved common factors.

Definition: factor model

$$X_i = \mu_i + \sum_{\ell=1}^{k} \lambda_{i\ell} F_\ell + U_i \quad (i = 1, \dots, p), \qquad \mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F} + \mathbf{U},$$

where $\boldsymbol{\Lambda} = (\lambda_{i\ell})$ is the $p \times k$ **loading matrix**, $\mathbf{F} \in \mathbb{R}^k$ the common factors and $\mathbf{U} \in \mathbb{R}^p$ the specific factors, with

$$\mathrm{E}(\mathbf{F}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{F}) = \mathbf{I}_k, \quad \mathrm{E}(\mathbf{U}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{U}) = \boldsymbol{\Psi} = \mathrm{diag}(\psi_1, \dots, \psi_p), \quad \mathrm{Cov}(\mathbf{F}, \mathbf{U}) = \mathbf{0}.$$

Theorem: covariance structure

$$\mathrm{E}(\mathbf{X}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}, \qquad \mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \boldsymbol{\Lambda},$$

$$\mathrm{Var}(X_i) = \underbrace{\sum_{\ell=1}^{k} \lambda_{i\ell}^2}_{h_i^2\text{: communality}} + \underbrace{\psi_i}_{\text{specific variance}}, \qquad \mathrm{Cov}(X_i, X_j) = \sum_{\ell} \lambda_{i\ell}\lambda_{j\ell} \ (i \neq j).$$

Proof. $\mathrm{Cov}(\boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}) = \boldsymbol{\Lambda}\mathrm{Cov}(\mathbf{F})\boldsymbol{\Lambda}^\top + \mathrm{Cov}(\mathbf{U}) + \boldsymbol{\Lambda}\mathrm{Cov}(\mathbf{F}, \mathbf{U}) + \mathrm{Cov}(\mathbf{U}, \mathbf{F})\boldsymbol{\Lambda}^\top = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$, and $\mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \mathrm{Cov}(\boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}, \mathbf{F}) = \boldsymbol{\Lambda}$. ■

Because $\boldsymbol{\Psi}$ is diagonal, covariances between variables come *entirely* from the common factors. The model explains the $p(p+1)/2$ entries of $\boldsymbol{\Sigma}$ with $pk + p$ parameters.

1.1 Non-identifiability and rotation

For orthogonal $\mathbf{T}$ $(k \times k)$, $\boldsymbol{\Lambda}^* = \boldsymbol{\Lambda}\mathbf{T}$ and $\mathbf{F}^* = \mathbf{T}^\top\mathbf{F}$ give $\boldsymbol{\Lambda}^*\mathbf{F}^* = \boldsymbol{\Lambda}\mathbf{F}$, $\mathrm{Cov}\mathbf{F}^* = \mathbf{I}_k$ and $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top$: loadings are determined only up to rotation. If $\hat{\boldsymbol{\Lambda}}$ is an MLE, so is $\hat{\boldsymbol{\Lambda}}\mathbf{R}$ for any orthogonal $\mathbf{R}$. Removing the $k(k-1)/2$ rotational degrees of freedom leaves $pk + p - k(k-1)/2$ parameters and

$$\nu = \frac{p(p+1)}{2} - \left\{pk + p - \frac{k(k-1)}{2}\right\} = \frac{(p-k)^2 - p - k}{2},$$

which must be positive for the model to impose a real restriction.

1.2 Invariance to scale changes

If $Y_i = c_i X_i$, i.e. $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\mathbf{X}$, then $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\boldsymbol{\mu} + \mathrm{diag}(\mathbf{c})\boldsymbol{\Lambda}\mathbf{F} + \mathrm{diag}(\mathbf{c})\mathbf{U}$ is again a factor model with $\boldsymbol{\Lambda}_Y = \mathrm{diag}(\mathbf{c})\boldsymbol{\Lambda}$ and $\boldsymbol{\Psi}_Y = \mathrm{diag}(\mathbf{c})^2\boldsymbol{\Psi}$. For standardised variables ($\mathrm{Var}X_i = 1$),

$$\lambda_{i\ell} = \mathrm{Cov}(X_i, F_\ell) = \mathrm{Corr}(X_i, F_\ell), \qquad \mathrm{Corr}(Y_i, F_\ell) = \lambda_{i\ell},$$

so loadings are read as variable–factor correlations. In practice factor analysis is often run on the correlation matrix $\mathbf{R}$.

2 Estimation

2.1 Principal component method

From $\widehat{\mathrm{Corr}}(\mathbf{X}) = \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} = \sum_{j=1}^{p} l_j \mathbf{u}_j \mathbf{u}_j^\top \approx \sum_{j=1}^{k} l_j \mathbf{u}_j \mathbf{u}_j^\top$ (largest k eigenvalues),

$$\hat{\mathbf{\Lambda}} = [\sqrt{l_1}\mathbf{u}_1, \ldots, \sqrt{l_k}\mathbf{u}_k], \qquad \hat{\mathbf{\Psi}} = \mathrm{diag}(\mathbf{R} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top).$$

The diagonal of $\mathbf{R}$ is reproduced exactly, and the sum of squared off-diagonal residuals of $\mathbf{R} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top - \hat{\mathbf{\Psi}}$ is at most $l_{k+1}^2 + \cdots + l_p^2$. Factor ℓ explains a proportion l_ℓ/p of the total variance. Since eigenvalues do not change, increasing k leaves earlier loadings unchanged.

Remark: number of factors with the principal component method

Heuristics are used when PCA is used to estimate the factor loadings:

1. use a **scree plot** or a **cumulative scree plot** of $l_1 \geq l_2 \geq \cdots \geq l_p$;
2. or use **Kaiser's rule**: retain factors $1, \ldots, k$ satisfying

$$l_k > \bar{l} = \frac{1}{p}\sum_{i=1}^{p} l_i.$$

When correlation PCA is used, $\sum_i l_i = \mathrm{tr}\mathbf{R} = p$, so $\bar{l} = 1$ and the rule keeps factors with $l_k > 1$ (a factor must explain more than one standardised variable's worth of variance).

These are the same rules as for choosing the number of principal components (Ch. 7); the likelihood ratio test of §2.4 is available only under the normal MLE approach.

2.2 Principal factor method

Starting from initial communality estimates $\hat{h}_i^2$ (e.g. $1 - 1/r^{ii}$, r^{ii} the diagonal of $\mathbf{R}^{-1}$), form the reduced correlation matrix $\mathbf{R} - \hat{\mathbf{\Psi}}$ and iterate

1. $\mathbf{R} - \hat{\mathbf{\Psi}} \approx \hat{\mathbf{\Lambda}}_{(k)}\hat{\mathbf{\Lambda}}_{(k)}^\top$ (top-k eigendecomposition of the reduced matrix);
2. $\hat{\psi}_{(k)i} = 1 - \sum_{\ell=1}^{k} \hat{\lambda}_{(k)i\ell}^2$

until convergence.

2.3 Maximum likelihood

Assume $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})$. Profiling $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, twice the negative log-likelihood is, up to constants and a factor n,

$$\ell(\hat{\boldsymbol{\mu}}, \mathbf{\Lambda}, \mathbf{\Psi}) = \log|\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi}| + \mathrm{tr}\{\mathbf{S}_0(\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})^{-1}\} + c.$$

For identifiability impose that $\mathbf{\Delta} = \mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda}$ is diagonal (if $\hat{\mathbf{\Lambda}}$ is an MLE, so is $\hat{\mathbf{\Lambda}}\mathbf{R}$).

Theorem: $\mathbf{\Lambda}$ given $\mathbf{\Psi}$

Given $\mathbf{\Psi} = \mathrm{diag}(\psi_1, \ldots, \psi_p)$, let $\mathbf{A} = \mathbf{\Psi}^{-1/2}\mathbf{S}_0\mathbf{\Psi}^{-1/2} = \sum_{i=1}^{p} d_i \mathbf{e}_i \mathbf{e}_i^\top$ ($d_1 \geq \cdots \geq d_p$) and $\delta_i = d_i - 1 > 0$ ($i \leq k$). Then

$$\hat{\mathbf{\Lambda}} = \mathbf{\Psi}^{1/2}[\mathbf{e}_1, \ldots, \mathbf{e}_k]\mathrm{diag}(\delta_1^{1/2}, \ldots, \delta_k^{1/2}).$$

Proof. Let $\boldsymbol{\Lambda}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}$ and $\boldsymbol{\Sigma}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Sigma}\boldsymbol{\Psi}^{-1/2} = \boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} + \mathbf{I}_p$. Then $\ell = \log|\boldsymbol{\Psi}| + \log|\boldsymbol{\Sigma}^*| + \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1})$. If $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top}$ has eigenvalues $\gamma_1 \geq \cdots \geq \gamma_k \geq 0 = \cdots = 0$, $\boldsymbol{\Sigma}^*$ has eigenvalues $1+\gamma_j$ and 1. Von Neumann's lower bound (pair the small eigenvalues of $\boldsymbol{\Sigma}^{*-1}$ with the large ones of $\mathbf{A}$) gives

$$\log|\boldsymbol{\Sigma}^*| + \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1}) \geq \sum_{j=1}^{k}\Big\{\log(1+\gamma_j) + \frac{d_j}{1+\gamma_j}\Big\} + \sum_{j>k} d_j,$$

with equality when the eigenvectors of $\boldsymbol{\Sigma}^*$ coincide with the $\mathbf{e}_j$ in the same order. Each term $\log x + d/x = \log d + \{d/x - \log(d/x)\} \geq \log d + 1$ (Lemma 3.3) is minimised at $x = d$, so $1+\gamma_j = d_j$, i.e. $\gamma_j = \delta_j$. Hence $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} = \sum_{j\leq k}\delta_j\mathbf{e}_j\mathbf{e}_j^\top$ and $\boldsymbol{\Lambda}^* = [\mathbf{e}_1,\ldots,\mathbf{e}_k]\operatorname{diag}(\delta_j^{1/2})$ up to rotation; this choice makes $\boldsymbol{\Lambda}^{*\top}\boldsymbol{\Lambda}^* = \boldsymbol{\Delta}$ diagonal. ∎

Theorem: $\boldsymbol{\Psi}$ given $\boldsymbol{\Lambda}$

The MLE satisfies $\operatorname{diag}(\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}) = \operatorname{diag}(\mathbf{S}_0)$ (the model reproduces the sample variances). Accordingly, given $\boldsymbol{\Lambda}$, update (handwritten notes)

$$\hat{\boldsymbol{\Psi}} = \operatorname{diag}(\hat{\psi}_1,\ldots,\hat{\psi}_p), \qquad \hat{\psi}_i = (\mathbf{S}_0 - \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top)_{ii}.$$

Alternate the two steps. If some $\hat{\psi}_i \leq 0$ (a Heywood case), bound it below by a small positive number.

2.4 Likelihood ratio test for the number of factors

$H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$ (k factors) versus H_1: unrestricted $\boldsymbol{\Sigma}$. With $L_1 = L(\bar{\mathbf{x}}, \mathbf{S}_0)$ and $L_0 = L(\bar{\mathbf{x}}, \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}})$,

$$W = -2\log\frac{L_0}{L_1} = n\log\Big(\frac{|\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}|}{|\mathbf{S}_0|}\Big) \;\dot{\sim}\; \chi^2_\nu,$$

$$\nu = \frac{p(p+1)}{2} - \Big\{pk + p - \frac{k(k-1)}{2}\Big\} = \frac{(p-k)^2 - p - k}{2}.$$

(At the MLE $\operatorname{tr}\{\mathbf{S}_0\hat{\boldsymbol{\Sigma}}^{-1}\} = p$, so the trace terms cancel.) With Bartlett's correction, reject the k-factor model if $\big(n - 1 - \frac{2p+4k+5}{6}\big)\ln\frac{|\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top+\hat{\boldsymbol{\Psi}}|}{|\mathbf{S}_0|} > \chi^2_\nu(\alpha)$.

3 Factor Rotation

Non-identifiability is used to find interpretable loadings, ideally a simple structure in which each variable loads heavily on only one factor. Rotate $\boldsymbol{\Lambda}^* = \boldsymbol{\Lambda}\mathbf{J}$ with $\mathbf{J}$ orthogonal.

Definition: varimax and quartimax

Let $\tilde{\lambda}^*_{i\ell} = \lambda^*_{i\ell}/h_i$ (normalised by the communality).

$$\text{Varimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{\ell=1}^{k} \widehat{\operatorname{Var}}\big((\tilde{\lambda}^*_{1\ell})^2,\ldots,(\tilde{\lambda}^*_{p\ell})^2\big)$$

— spread the squared loadings within each *column* (factor), so each factor has some large and some near-zero loadings.

$$\text{Quartimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{i=1}^{p} \widehat{\operatorname{Var}}\big((\lambda^*_{i1})^2,\ldots,(\lambda^*_{ik})^2\big)$$

— spread the squared loadings within each *row* (variable), so each variable concentrates on one factor.

Rotation leaves the communalities h_i^2 and $\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top$ unchanged.

4 Factor Score Prediction

4.1 Weighted least squares (Bartlett) scores

(1) Choose $\hat{\mathbf{F}}$ to minimise the distance between $\mathbf{X}$ and $\boldsymbol{\mu} + \mathbf{\Lambda F}$: with the estimates fixed, regard $\mathbf{x}_i - \bar{\mathbf{x}} = \hat{\mathbf{\Lambda}}\mathbf{F}_i + \mathbf{U}_i$ as a heteroscedastic regression,

$$\hat{\mathbf{F}}_i^{\text{wls}} = \arg\min_{\mathbf{F}_i} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\mathbf{\Lambda}}\mathbf{F}_i)^\top \hat{\mathbf{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\mathbf{\Lambda}}\mathbf{F}_i) = (\hat{\mathbf{\Lambda}}^\top \hat{\mathbf{\Psi}}^{-1} \hat{\mathbf{\Lambda}})^{-1} \hat{\mathbf{\Lambda}}^\top \hat{\mathbf{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$$

(set the gradient of the quadratic form to zero).

4.2 Regression scores

(2) Since $\begin{pmatrix} \mathbf{X} \\ \mathbf{F} \end{pmatrix} \sim N_{p+k}\Big(\begin{pmatrix} \boldsymbol{\mu} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{\Lambda\Lambda}^\top + \mathbf{\Psi} & \mathbf{\Lambda} \\ \mathbf{\Lambda}^\top & \mathbf{I}_k \end{pmatrix} \Big)$, the conditional expectation (Result 4.6) gives

$$\hat{\mathbf{F}}_i^{\text{reg}} = \mathrm{E}(\mathbf{F} \mid \mathbf{X} = \mathbf{x}_i) = \hat{\mathbf{\Lambda}}^\top (\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}})^{-1} (\mathbf{x}_i - \bar{\mathbf{x}}).$$

Theorem: relation between the two scores

With $\mathbf{\Delta} = \mathbf{\Lambda}^\top \mathbf{\Psi}^{-1} \mathbf{\Lambda}$, $\hat{\mathbf{F}}^{\text{reg}} = (\mathbf{I} + \mathbf{\Delta})^{-1} \mathbf{\Delta}\, \hat{\mathbf{F}}^{\text{wls}} = (\mathbf{I} + \mathbf{\Delta}^{-1})^{-1} \hat{\mathbf{F}}^{\text{wls}}$. If $\mathbf{\Delta}$ is diagonal each component is shrunk by $\delta_\ell/(1+\delta_\ell) < 1$, so $|\hat{F}_\ell^{\text{wls}}| \geq |\hat{F}_\ell^{\text{reg}}|$.

Proof. By Woodbury, $(\mathbf{\Lambda\Lambda}^\top + \mathbf{\Psi})^{-1} = \mathbf{\Psi}^{-1} - \mathbf{\Psi}^{-1}\mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta})^{-1}\mathbf{\Lambda}^\top \mathbf{\Psi}^{-1}$, so

$$\mathbf{\Lambda}^\top (\mathbf{\Lambda\Lambda}^\top + \mathbf{\Psi})^{-1} = \{\mathbf{I} - \mathbf{\Delta}(\mathbf{I} + \mathbf{\Delta})^{-1}\} \mathbf{\Lambda}^\top \mathbf{\Psi}^{-1} = (\mathbf{I} + \mathbf{\Delta})^{-1} \mathbf{\Lambda}^\top \mathbf{\Psi}^{-1}$$

(as $\mathbf{I} - \mathbf{\Delta}(\mathbf{I} + \mathbf{\Delta})^{-1} = (\mathbf{I} + \mathbf{\Delta} - \mathbf{\Delta})(\mathbf{I} + \mathbf{\Delta})^{-1}$), while $\mathbf{\Lambda}^\top \mathbf{\Psi}^{-1}(\mathbf{x} - \bar{\mathbf{x}}) = \mathbf{\Delta}\hat{\mathbf{F}}^{\text{wls}}$. ■

Regression scores shrink towards 0, reflecting the prior $\mathbf{F} \sim N(\mathbf{0}, \mathbf{I})$ (a Bayes estimate); WLS scores treat $\mathbf{F}$ as a fixed parameter and are conditionally unbiased.

5 PCA versus Factor Analysis

	PCA	Factor analysis
Goal	linear combinations explaining total variance $\mathrm{tr}\mathbf{\Sigma}$	latent factors explaining covariances (off-diagonal)
Model	none (descriptive transformation)	$\mathbf{\Sigma} = \mathbf{\Lambda\Lambda}^\top + \mathbf{\Psi}$
Direction	variables $\to$ components ($\mathbf{Z} = \mathbf{U}^\top \mathbf{X}$)	factors $\to$ variables ($\mathbf{X} = \boldsymbol{\mu} + \mathbf{\Lambda F} + \mathbf{U}$)
Uniqueness	unique if eigenvalues distinct	only up to rotation
Increasing k	earlier components unchanged	(MLE) earlier loadings change

Chapter 10. Discrimination and Classification

Bayes rule, LDA/QDA, Fisher's discriminant, logistic regression, assessing classifiers

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1 The Bayes Classification Rule

1.1 Setting

Let $Y \in \{1, \dots, K\}$ be the group and $\mathbf{X} \in \mathbb{R}^p$ the features, with priors $P(Y = k) = \pi_k$ and $\mathbf{X} \mid Y = k \sim f_k$. For $K = 2$: $\mathbf{X} \mid Y = 1 \sim f_1(\mathbf{x})$, $\mathbf{X} \mid Y = 2 \sim f_2(\mathbf{x})$, $P(Y = 1) = \pi_1$, and

$$\eta(\mathbf{x}) := P(Y = 1 \mid \mathbf{X} = \mathbf{x}) = \frac{f_1(\mathbf{x})\pi_1}{f_1(\mathbf{x})\pi_1 + f_2(\mathbf{x})\pi_2}.$$

The misclassification probability of a classifier δ (or ϕ) : $\mathbb{R}^p \to \{1, \dots, K\}$ is $P(\phi(\mathbf{X}) \neq Y) = \sum_k P(\phi(\mathbf{X}) \neq k \mid Y = k)P(Y = k)$.

Theorem: optimality of the Bayes rule

$\phi_{\text{Bayes}}(\mathbf{x}) = \arg\max_k P(Y = k \mid \mathbf{X} = \mathbf{x}) = \arg\max_k \pi_k f_k(\mathbf{x})$ minimises the misclassification probability: $P(\phi_{\text{Bayes}}(\mathbf{X}) \neq Y) \leq P(\phi(\mathbf{X}) \neq Y)$ for every ϕ.

$K = 2$ **(handwritten notes).**

$$\begin{aligned}
\text{Risk}(\delta) &= \mathrm{E}\big[\mathbf{1}_{\{Y \neq \delta(\mathbf{X})\}}\big] = \mathrm{E}_Y\Big[\mathrm{E}_{\mathbf{X}|Y}\big[\mathbf{1}_{\{Y \neq \delta(\mathbf{X})\}} \mid Y\big]\Big] \\
&= \pi_1 \mathrm{E}_{\mathbf{X}|Y=1}\big[\mathbf{1}_{\{\delta(\mathbf{X}) \neq 1\}} \mid Y = 1\big] + \pi_2 \mathrm{E}_{\mathbf{X}|Y=2}\big[\mathbf{1}_{\{\delta(\mathbf{X}) \neq 2\}} \mid Y = 2\big] \\
&= \int \underbrace{\Big[\pi_1 \mathbf{1}_{\{\delta(\mathbf{x}) \neq 1\}} f_1(\mathbf{x}) + \pi_2 \mathbf{1}_{\{\delta(\mathbf{x}) \neq 2\}} f_2(\mathbf{x})\Big]}_{=:h(\mathbf{x})} d\mathbf{x}.
\end{aligned}$$

At each $\mathbf{x}$, $\delta(\mathbf{x}) = 1$ gives $h(\mathbf{x}) = \pi_2 f_2(\mathbf{x})$ and $\delta(\mathbf{x}) = 2$ gives $h(\mathbf{x}) = \pi_1 f_1(\mathbf{x})$. To minimise the integral choose the smaller pointwise: if $\pi_2 f_2(\mathbf{x}) < \pi_1 f_1(\mathbf{x})$ choose $\delta(\mathbf{x}) = 1$; if $>$, choose $\delta(\mathbf{x}) = 2$. Thus $\phi_{\text{Bayes}}(\mathbf{x}) = \arg\max_k P(Y = k \mid \mathbf{X} = \mathbf{x})$. For general K the same argument minimises $1 - P(Y = \phi(\mathbf{x}) \mid \mathbf{x})$ pointwise. ■

Remark: misclassification costs

If $c(2 \mid 1)$ is the cost of classifying a member of group 1 into group 2, the rule minimising expected cost classifies into group 1 when $\dfrac{f_1(\mathbf{x})}{f_2(\mathbf{x})} \geq \dfrac{c(1 \mid 2)}{c(2 \mid 1)} \dfrac{\pi_2}{\pi_1}$.

2 Normal Populations: LDA and QDA

2.1 Common covariance: linear discriminant analysis

If $f_k = N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$ ($\boldsymbol{\Sigma}_k = \boldsymbol{\Sigma}$), then $\log\{\pi_k f_k(\mathbf{x})\} = -\frac{1}{2}\mathbf{x}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \log \pi_k +$ const, and the first term does not depend on k, so

$$\phi(\mathbf{x}) = \arg\min_k \delta_k(\mathbf{x}), \qquad \delta_k(\mathbf{x}) = b_{0k} - \mathbf{b}_k^\top \mathbf{x}, \quad b_{0k} = \tfrac{1}{2}\boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k - \log(\pi_k), \quad \mathbf{b}_k = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k.$$

The discriminant function is linear in $\mathbf{x}$: **linear discriminant analysis (LDA)**. For $K = 2$, $\delta_1 < \delta_2$ rearranges to

$$\phi(\mathbf{x}) = 1 \;\text{ if }\; \mathbf{b}^\top\Big(\mathbf{x} - \frac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2}{2}\Big) < \log\Big(\frac{\pi_1}{\pi_2}\Big), \qquad \mathbf{b} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1).$$

$K = 2$. $\delta_1 < \delta_2 \iff (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - \frac{1}{2}(\boldsymbol{\mu}_1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2) > \log(\pi_2/\pi_1)$. Since $\boldsymbol{\mu}_1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)$, this is $-\mathbf{b}^\top(\mathbf{x} - \frac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2}{2}) > \log(\pi_2/\pi_1)$. ■

The sample version plugs in $\boldsymbol{\mu}_k \to \bar{\mathbf{x}}_k$, $\boldsymbol{\Sigma} \to \mathbf{S}_p$ (pooled covariance), $\pi_k \to n_k/n$: $\phi(\mathbf{x}) = \arg\max_k (b_{0k} - \mathbf{b}_k^\top \mathbf{x})$-type rule with $\hat{\mathbf{b}}_k = \mathbf{S}_p^{-1}\bar{\mathbf{x}}_k$, and for $K = 2$, $\hat{\mathbf{b}} = \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$.

2.2 Different covariances: quadratic discriminant analysis

If $f_k = N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$,

$$\phi(\mathbf{x}) = \arg\max_k \Big\{ -\tfrac{1}{2}\log|\boldsymbol{\Sigma}_k| - \tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1}(\mathbf{x}-\boldsymbol{\mu}_k) + \log \pi_k \Big\} = \arg\max_k \left(b_{0k} - \mathbf{b}_k^\top \mathbf{x} + \mathbf{x}^\top \mathbf{C}_k \mathbf{x}\right)$$

with $\mathbf{C}_k = -\frac{1}{2}\boldsymbol{\Sigma}_k^{-1}$. The boundaries are quadratic surfaces: **QDA**. It needs $K \cdot p(p+1)/2$ more parameters, so with small samples LDA is often more stable (bias–variance trade-off).

3 Fisher's Linear Discriminant

Without distributional assumptions, find the projection that best separates the groups.

Definition: Fisher's LDA direction ($K = 2$)

$$\mathbf{u}_0 = \arg\max_{\mathbf{u}\in\mathbb{R}^p} \frac{(\mathbf{u}^\top \bar{\mathbf{x}}_1 - \mathbf{u}^\top \bar{\mathbf{x}}_2)^2}{\mathbf{u}^\top \mathbf{S}_p \mathbf{u}} :$$

squared difference of projected means over projected within-group variance.

Theorem: solution

$\mathbf{u}_0 \propto \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)$ (equivalently $\mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$), and the maximum is $\mathbf{d}^\top \mathbf{S}_p^{-1}\mathbf{d}$ with $\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$, the squared sample Mahalanobis distance.

Proof. In general, to maximise $\frac{\mathbf{u}^\top \mathbf{A}\mathbf{u}}{\mathbf{u}^\top \mathbf{B}\mathbf{u}}$ ($\mathbf{B} \succ 0$) use $L(\mathbf{u}, \lambda) = \mathbf{u}^\top \mathbf{A}\mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{B}\mathbf{u} - 1)$: $\partial L/\partial \mathbf{u} = \mathbf{0} \Rightarrow \mathbf{A}\mathbf{u} = \lambda \mathbf{B}\mathbf{u} \Rightarrow \mathbf{B}^{-1}\mathbf{A}\mathbf{u} = \lambda \mathbf{u}$, so the top eigenvector of $\mathbf{B}^{-1}\mathbf{A}$ is optimal. With $\mathbf{A} = \mathbf{d}\mathbf{d}^\top$ (rank 1) and $\mathbf{B} = \mathbf{S}_p$, $\mathbf{S}_p^{-1}\mathbf{d}\mathbf{d}^\top$ has a single nonzero eigenvalue $\mathbf{d}^\top \mathbf{S}_p^{-1}\mathbf{d}$ with unique eigenvector $\mathbf{S}_p^{-1}\mathbf{d}$ (rank 1). Directly: with $\mathbf{v} = \mathbf{S}_p^{1/2}\mathbf{u}$, Cauchy–Schwarz gives $(\mathbf{u}^\top \mathbf{d})^2/\mathbf{u}^\top \mathbf{S}_p \mathbf{u} = (\mathbf{v}^\top \mathbf{S}_p^{-1/2}\mathbf{d})^2/\mathbf{v}^\top \mathbf{v} \le \mathbf{d}^\top \mathbf{S}_p^{-1}\mathbf{d}$. ■

Remark: Fisher's rule = LDA with equal priors

With the discriminant score $y_i = \mathbf{b}_Y^\top \mathbf{x}_i$, $\mathbf{b}_Y = \mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$, and the cut-off $b_{0Y} = \mathbf{b}_Y^\top(\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2)/2$,

$$\text{“}\mathbf{b}_Y^\top \mathbf{y} - b_{0Y} < 0 \Rightarrow 1\text{”} \iff \text{“}\left[\mathbf{S}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)\right]^\top \left(\mathbf{x} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2}\right) < 0 \Rightarrow 1\text{”},$$

which is sample LDA with $\pi_1 = \pi_2$. A rule derived without normality coincides with the Bayes rule under normality.

Several groups ($K \ge 3$): $\max_{\mathbf{u}} \dfrac{\mathbf{u}^\top \mathbf{B}\mathbf{u}}{\mathbf{u}^\top \mathbf{W}\mathbf{u}}$ ($\mathbf{B}, \mathbf{W}$ the MANOVA between/within SSP matrices) is solved by the eigenvectors of $\mathbf{W}^{-1}\mathbf{B}$; there are at most $\min(p, K-1)$ nonzero eigenvalues, the same η_i that appear in the Wilks/Pillai statistics of Ch. 6.

4 Logistic Regression

For $K = 2$, $Y \in \{0, 1\}$, model the posterior directly:

$$\operatorname{logit}(\eta(\mathbf{x})) := \log\Big(\frac{\eta(\mathbf{x})}{1-\eta(\mathbf{x})}\Big) = b + \boldsymbol{\beta}^\top \mathbf{x}, \qquad \eta(\mathbf{x}) = P(Y = 1 \mid \mathbf{X} = \mathbf{x}).$$

Classification: "$\eta(\mathbf{x}) > \frac{1}{2} \Rightarrow 1$" $\iff$ "$f(\mathbf{x}) = b + \boldsymbol{\beta}^\top \mathbf{x} > 0 \Rightarrow 1$" (a linear boundary). The log-likelihood is

$$\begin{aligned}\ell(b, \boldsymbol{\beta}) &= \sum_{i=1}^{n} \left[y_i \log(\eta(\mathbf{x}_i; b, \boldsymbol{\beta})) + (1 - y_i) \log\{1 - \eta(\mathbf{x}_i; b, \boldsymbol{\beta})\} \right] \\ &= \sum_{i=1}^{n} \left[y_i (b + \boldsymbol{\beta}^\top \mathbf{x}_i) - \log\{1 + \exp(b + \boldsymbol{\beta}^\top \mathbf{x}_i)\} \right],\end{aligned}$$

which is concave and maximised by Newton–Raphson (IRLS). In the equal-covariance normal model $\log \frac{\pi_1 f_1}{\pi_0 f_0}$ is exactly linear in $\mathbf{x}$, so the logistic model holds; LDA models the joint distribution, logistic regression only the conditional one.

Derivation: the second equality and the derivatives

$\log \eta = (b + \boldsymbol{\beta}^\top \mathbf{x}) - \log(1 + e^{b+\boldsymbol{\beta}^\top \mathbf{x}})$ and $\log(1 - \eta) = -\log(1 + e^{b+\boldsymbol{\beta}^\top \mathbf{x}})$, so $y \log \eta + (1 - y) \log(1 - \eta) = y(b + \boldsymbol{\beta}^\top \mathbf{x}) - \log(1 + e^{b+\boldsymbol{\beta}^\top \mathbf{x}})$. The gradient is $\sum_i (y_i - \eta_i)(1, \mathbf{x}_i^\top)^\top$ and the Hessian $-\sum_i \eta_i(1 - \eta_i)(1, \mathbf{x}_i^\top)^\top (1, \mathbf{x}_i^\top) \preceq 0$.

5 Simple Classifiers in High Dimensions

If $p > n$, $\mathbf{S}_p$ is singular and LDA cannot be used (e.g. the Golub data).

- **Nearest centroid**: LDA with $\boldsymbol{\Sigma} = \mathbf{I}$. $\phi(\mathbf{x}) = 1$ if $\mathbf{b}^\top (\mathbf{x} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2}) < 0$, $\mathbf{b} = \bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1$ (assign to the closer mean).
- **Naive Bayes classifier (diagonal LDA)**: assume independent variables and replace $\mathbf{S}_p$ by $\mathbf{D}_p = \operatorname{diag}(\mathbf{S}_p)$: $\phi(\mathbf{x}) = 1$ if $\mathbf{b}^\top (\mathbf{x} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2}) < 0$, $\mathbf{b} = \mathbf{D}_p^{-1}(\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1)$.
- **k-nearest neighbours**: majority vote among the k training points closest to $\mathbf{x}$; non-parametric, high variance for small k and high bias for large k.

6 Assessment of Classifiers

- True error rate $P(\phi(\mathbf{X}) \neq Y) = \sum_{k=1}^{K} P(\phi(\mathbf{X}) \neq k \mid Y = k) P(Y = k)$; ϕ is optimal if $P(\phi(\mathbf{X}) \neq Y) \leq P(\varphi(\mathbf{X}) \neq Y)$ for all φ.
- Empirical misclassification rate $\hat{P}(\hat{\phi}_D(\mathbf{X}) \neq Y \mid D) = \frac{1}{M} \sum_{i=1}^{M} \mathbf{1}_{\{\hat{\phi}(\mathbf{x}_i) \neq y_i\}}$.
- **Training and test sets**: build $\hat{\phi}$ from $\mathbf{X}_{\text{tr}}$ and estimate the misclassification rate on $\mathbf{X}_{\text{test}}$. The training error (APER) is optimistically biased.
- **Confusion matrix**: actual group by predicted group (training and testing confusion matrices).
- **K-fold cross-validation**: split into K parts, use each in turn as the test set and average. $K \uparrow \Rightarrow$ bias $\downarrow$, variance $\uparrow$; $K \downarrow \Rightarrow$ the reverse. $K = n$ is LOOCV (Lachenbruch's holdout).

Chapter 11. Cluster Analysis

K-means, K-medoids, hierarchical clustering, Gaussian mixtures and the EM algorithm, choosing K

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Unlike classification (Ch. 10), clustering groups similar observations without labels (unsupervised learning). A clustering is an encoder $C : \{1, \dots, n\} \to \{1, \dots, K\}$.

1 Dissimilarity and Within-Cluster Scatter

Definition: within-cluster scatter

For a dissimilarity $d(i, j) \geq 0$,

$$W(C) = \frac{1}{2} \sum_{k=1}^{K} \frac{1}{n_k} \sum_{i:C(i)=k} \sum_{j:C(j)=k} d(i, j), \qquad n_k = \#\{i : C(i) = k\}.$$

Smaller W is better.

Theorem: with squared Euclidean distance $W(C)$ is the within-cluster sum of squares

If $d(i, j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2$, then

$$W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|_2^2, \qquad \bar{\mathbf{x}}_{(k)} = \frac{1}{n_k} \sum_{C(i)=k} \mathbf{x}_i,$$

and $T = \sum_i \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2 = W(C) + B(C)$ with $B(C) = \sum_k n_k \|\bar{\mathbf{x}}_{(k)} - \bar{\mathbf{x}}\|^2$.

Proof. Within cluster k, $\mathbf{x}_i - \mathbf{x}_j = (\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}) - (\mathbf{x}_j - \bar{\mathbf{x}}_{(k)})$ and $\sum_{C(i)=k} (\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}) = \mathbf{0}$, so

$$\sum_{C(i)=k} \sum_{C(j)=k} \|\mathbf{x}_i - \mathbf{x}_j\|^2 = 2n_k \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2;$$

multiply by $\frac{1}{2} \cdot \frac{1}{n_k}$. $T = W + B$ is the trace of the MANOVA SSP decomposition (Ch. 6). ∎

Since T is fixed, minimising W is maximising B.

1.1 Limits of combinatorial algorithms

The number of ways to partition n items into K non-empty clusters (Stirling number of the second kind) is

$$A(n, K) = \frac{1}{K!} \sum_{k=1}^{K} (-1)^{K-k} \binom{K}{k} k^n \sim \frac{K^n}{K!}.$$

E.g. $A(19, 4) \approx 10^{10}$: exhaustive search is impossible, so greedy iterative algorithms are used.

2 K-means and K-medoids

2.1 K-means

With $d(i, j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2$, any point $\boldsymbol{m}$ satisfies $\sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{m}\|^2 \geq \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2$, so

$$W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|_2^2 = \min_{\boldsymbol{m}_1, \dots, \boldsymbol{m}_K} L(C \mid \boldsymbol{m}), \qquad L(C \mid \boldsymbol{m}) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{m}_k\|_2^2.$$

Definition: K-means algorithm

(i) Minimise over C with $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$ fixed: update $C(i) = k$ for the closest centre.

(ii) Minimise over $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$ with C fixed: update $\boldsymbol{m}_k$ by the new average of cluster k.

(iii) Iterate (i), (ii) until $L(C \mid \boldsymbol{m})$ does not change.

Theorem: convergence

K-means stops after finitely many iterations at a local minimum.

Proof. Each step minimises L in one block of variables with the other fixed, so L is non-increasing. There are finitely many encoders C and step (ii) is unique for a given C, so while L strictly decreases no C can recur; the algorithm stops in finitely many steps. ■

The result depends on the starting values, so run several starts and keep the smallest W. K-means is scale-dependent; consider standardising.

2.2 K-medoids

Use an arbitrary dissimilarity and restrict centres to observations.

(i) Minimise over C: update $C(i) = k$ for the closest medoid $\boldsymbol{m}_k$.

(ii) Minimise over $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$: update $\boldsymbol{m}_k$ by the item in the kth cluster that minimises the total dissimilarity to the cluster.

(iii) Iterate (i), (ii) until $L(C \mid \boldsymbol{m})$ does not change.

Using actual observations as representatives makes it robust to outliers and usable for categorical data.

3 Hierarchical Clustering

3.1 Agglomerative

Start with singletons and repeatedly merge the two closest clusters. Between-cluster distances (linkage):

Single linkage	$d(G_1, G_2) = \min_{i \in G_1, j \in G_2} d_{ij}$
Complete linkage	$d(G_1, G_2) = \max_{i \in G_1, j \in G_2} d_{ij}$
Average linkage	$d(G_1, G_2) = \text{average}_{i \in G_1, j \in G_2} d_{ij}$

Single linkage tends to produce long chained clusters, complete linkage compact round ones. The result is drawn as a dendrogram and cut at a chosen height.

Distances between observations:

$$p\text{-norm: } d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_p = \Big(\sum_{l=1}^{d} |x_l - y_l|^p \Big)^{1/p},$$

$$\text{standardised distance: } d^2(\mathbf{x}, \mathbf{y}) = \sum_{l=1}^{d} \frac{(x_l - y_l)^2}{s_l^2} = (\mathbf{x} - \mathbf{y})^\top \mathbf{D}^{-1} (\mathbf{x} - \mathbf{y}).$$

3.2 Divisive

Start from a single cluster and split repeatedly (e.g. $K = 2$ means at each step).

4 Gaussian Mixtures and the EM Algorithm

4.1 Model

$\mathbf{X}_1, \dots, \mathbf{X}_n \overset{\text{iid}}{\sim} g(\mathbf{x})$ with latent labels $\delta_1, \dots, \delta_n \overset{\text{iid}}{\sim}$, $P(\delta_i = k) = \pi_k$, and $\mathbf{X} \mid \delta = k \sim N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$:

$$\mathbf{X} \sim g(\mathbf{x}) = \sum_{k=1}^{K} \operatorname{pdf}(\mathbf{x} \mid \delta = k) P(\delta = k) = \sum_{k=1}^{K} \pi_k g_k(\mathbf{x}),$$

$$L(\boldsymbol{\theta} \mid \mathbf{X}) = \prod_{i=1}^{n} g(\mathbf{x}_i) = \prod_{i=1}^{n} \Big\{ \sum_{k=1}^{K} g_k(\mathbf{x}_i) \pi_k \Big\},$$

giving $\hat{\pi}_k, \hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k$. Then $\hat{P}(\delta = k \mid \mathbf{x}) = \hat{g}_k(\mathbf{x}) \hat{P}(\delta = k) / \hat{g}(\mathbf{x})$ and $C : \mathcal{X} \to \{1, \dots, K\}$, $C(\mathbf{x}) = k$ iff $\hat{P}(\delta = k \mid \mathbf{x}) \geq \hat{P}(\delta = j \mid \mathbf{x})$ for all j (soft clustering). With all $\boldsymbol{\Sigma}_k = \sigma^2 \mathbf{I}$ and $\sigma \to 0$ this becomes K-means.

4.2 The EM algorithm ($K = 2$)

$\mathbf{X}_1, \dots, \mathbf{X}_n \sim \pi_1 \phi(\mathbf{x}; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \pi_0 \phi(\mathbf{x}; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, $\pi_0 = 1 - \pi_1$. The observed-data log-likelihood

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log \{ \pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) + \pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) \}$$

has no closed-form maximiser (log of a sum). If $\delta_1, \dots, \delta_n \in \{0, 1\}$ were observed, $L(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n) = \prod_i \operatorname{pdf}(\mathbf{x}_i, \delta_i \mid \boldsymbol{\theta}) = \prod_i \operatorname{pdf}(\mathbf{x}_i \mid \delta_i, \boldsymbol{\theta}) \operatorname{pdf}(\delta_i)$ and

$$\ell(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n) = \sum_{i : \delta_i = 1} \log \pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \sum_{i : \delta_i = 0} \log \pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$$

$$= \sum_{i=1}^{n} \left[\delta_i \log \pi_1 \phi_1(\mathbf{x}_i) + (1 - \delta_i) \log \pi_0 \phi_0(\mathbf{x}_i) \right],$$

which separates and gives closed forms such as $\hat{\pi}_1^{\text{MLE}} = n_1 / n$.

Definition: EM algorithm

(i) $\ell(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n)$ has an easy MLE, but $\delta_1, \dots, \delta_n$ are unobserved; (ii) iteratively approximate the maximum. At $\boldsymbol{\theta}^{(m)}$:

- **E-step**: predict δ_i by $\hat{\delta}_i = \mathrm{E}(\delta_i \mid \mathbf{x}_i, \boldsymbol{\theta}^{(m)})$, i.e. compute $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) := \mathrm{E}[\ell(\boldsymbol{\theta} \mid D) \mid D_{\text{obs}}, \boldsymbol{\theta}^{(m)}]$. Since the complete-data log-likelihood is linear in δ_i,

$$\hat{\delta}_i = \frac{\pi_1^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_1^{(m)}, \boldsymbol{\Sigma}_1^{(m)})}{\pi_0^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_0^{(m)}, \boldsymbol{\Sigma}_0^{(m)}) + \pi_1^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_1^{(m)}, \boldsymbol{\Sigma}_1^{(m)})} \quad \text{(responsibility)}.$$

- **M-step**: $\boldsymbol{\theta}^{(m+1)} = \arg\max_{\boldsymbol{\theta}} L(\boldsymbol{\theta} \mid \mathbf{X}^n, \hat{\boldsymbol{\delta}}^n)$, i.e. $\arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$:

$$\pi_1 = \frac{\sum \hat{\delta}_i}{n}, \quad \boldsymbol{\mu}_1 = \frac{\sum \hat{\delta}_i \mathbf{x}_i}{\sum \hat{\delta}_i}, \quad \boldsymbol{\Sigma}_1 = \frac{\sum \hat{\delta}_i (\mathbf{x}_i - \boldsymbol{\mu}_1)(\mathbf{x}_i - \boldsymbol{\mu}_1)^\top}{\sum \hat{\delta}_i},$$

and similarly for component 0 with weights $1 - \hat{\delta}_i$.

Repeat until $d(\boldsymbol{\theta}^{(m)}, \boldsymbol{\theta}^{(m+1)})$ is small.

Theorem: monotonicity of EM

$\ell(\boldsymbol{\theta}^{(m+1)}) \geq \ell(\boldsymbol{\theta}^{(m)})$.

Proof. $\ell(\boldsymbol{\theta}) = Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$ with $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \mathrm{E}[\log p(\boldsymbol{\delta} \mid \mathbf{X}, \boldsymbol{\theta}) \mid \mathbf{X}, \boldsymbol{\theta}^{(m)}]$. By Gibbs' (Jensen's) inequality $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) \leq H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})$ (KL divergence ≥ 0). The M-step does not decrease Q, so $\ell(\boldsymbol{\theta}^{(m+1)}) - \ell(\boldsymbol{\theta}^{(m)}) = \{Q(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) - Q(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})\} - \{H(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})\} \geq 0$. ■

EM may converge to a local maximum (or saddle point), and the likelihood is unbounded as some $\boldsymbol{\Sigma}_k$ becomes singular, so use several starts and regularisation.

5 Choosing K and Other Topics

- **Scatter decomposition**: $W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2$, $B(C) = \sum_{k=1}^{K} n_k \|\bar{\mathbf{x}}_{(k)} - \bar{\mathbf{x}}\|^2$. W always decreases as K grows, so it cannot choose K by itself (elbow method: where the decrease flattens).
- Indices: $\mathrm{CI}(K, c) = W(K, c)/T$ (smaller is better) and Calinski–Harabasz $\mathrm{CH}(K, c) = \dfrac{B(K, c)/(K-1)}{W(K, c)/(n-K)}$ (larger is better; the analogue of the one-way MANOVA F).
- **Gap statistic**: compare with the expected $\log W_K^*$ under a reference distribution (uniform over the data range),

$$\mathrm{Gap}(K) = \mathrm{E}^*[\log W_K^*] - \log W_K, \qquad s(K) = \text{(adjusted) s.d. of } \log W_K^*,$$

  and choose the smallest K with $\mathrm{Gap}(K) \geq \mathrm{Gap}(K+1) - s(K+1)$.
- **Clustering variables**: define $d(v_i, v_j) = 1 - r_{ij}$ or $1 - r_{ij}^2$ and apply hierarchical clustering.